

Name: Viswapriya S
Course: Prompt Engineering
Slot: T2-H16

Content Creation

Using Prompt Patterns

How different prompt structures affect quality, coherence, creativity, and structure of AI-generated content

Query Decomposition

Decision-Making

Semantic Filtering

Structured Prompting

Creative Prompting

Assignment Objectives

01

Create Prompts

Design diverse prompt structures for content generation tasks across multiple domains.

02

Generate Content

Produce reports, articles, case studies, and creative content using AI tools.

03

Compare Outputs

Systematically compare results from different prompt styles side-by-side.

04

Analyze Effectiveness

Evaluate and critique prompt engineering techniques for quality and coherence.

5 Prompt Engineering Techniques

Each technique shapes AI output in fundamentally different ways

1



Query Decomposition

Breaking complex questions into structured sub-questions

2



Decision-Making Prompts

Guiding AI through structured reasoning and trade-offs

3



Semantic Filtering

Constraining responses by meaning, tone, and relevance

4



Structured Prompting

Using templates, roles, and format instructions

5



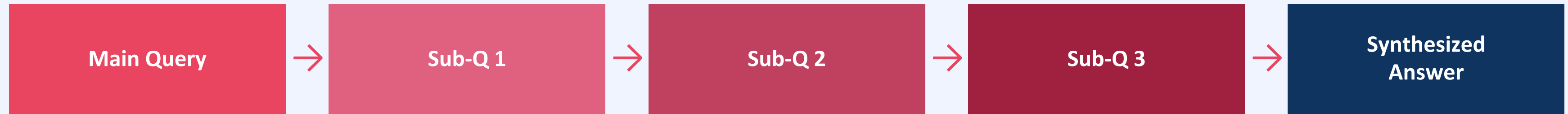
Creative Prompting

Unlocking imaginative, non-linear AI responses

Query Decomposition

What it is:

Breaking a complex question into a series of smaller, manageable sub-questions that guide the AI step by step.



Example:

Complex Query: "Write a report on climate change impact on agriculture"

Decomposed: 1) What are climate change effects? | 2) How does it affect crop yield? | 3) Which regions are most impacted? | 4) What are mitigation strategies?

Decision-Making Prompts

How It Works

- 1 Define the decision context clearly
- 2 List available options or constraints
- 3 Ask AI to evaluate trade-offs
- 4 Request a justified recommendation
- 5 Validate reasoning for coherence

Prompt Template

"You are an expert advisor. Given [context], evaluate the following options: [A, B, C]. Consider [criteria]. Recommend the best option with reasoning."

Why It Works

Structured reasoning forces the AI to consider multiple perspectives, reduces hallucinations, and produces more balanced, justified conclusions — ideal for case studies and analytical reports.

Semantic Filtering

Tone Filtering

Academic / Formal

Casual / Conversational

Persuasive / Marketing

Neutral / Journalistic

Scope Filtering

Limit to specific domain

Exclude irrelevant details

Focus on key variables

Restrict time period

Quality Filtering

Request verified facts only

Avoid speculation

Use reliable sources

Flag uncertain claims

💡 Tip: Combine multiple filters — e.g., "Write in an academic tone, focused only on post-2020 data, avoiding speculation."

Structured Prompting

The ROLE-TASK-FORMAT Template

ROLE

You are a senior data scientist...

CONTEXT

Given this dataset on customer churn...

TASK

Analyze trends and identify key drivers...

FORMAT

Structure as: Summary → Findings → Recommendations

CONSTRAINTS

Use plain English. Max 300 words. Cite evidence.

Key Benefits



Consistency:

Every output follows the same quality structure



Reproducibility:

Same template → predictable results



Scalability:

Reuse templates for batch content creation



Error Reduction:

Fewer vague or off-topic responses



Comparability:

Easy side-by-side output analysis

Creative Prompting

Persona Assignment

"Write as a futurist in 2075 looking back at how AI changed education."

Narrative Framing

"Tell the story of climate change through the eyes of a glacier."

Constraint-Based

"Explain blockchain in exactly 6 sentences using no technical jargon."

What-If Scenarios

"What if social media had been invented in the 1800s? Write an article."

Creative prompting unlocks imagination — best for storytelling, marketing copy, brainstorming, and exploratory articles.

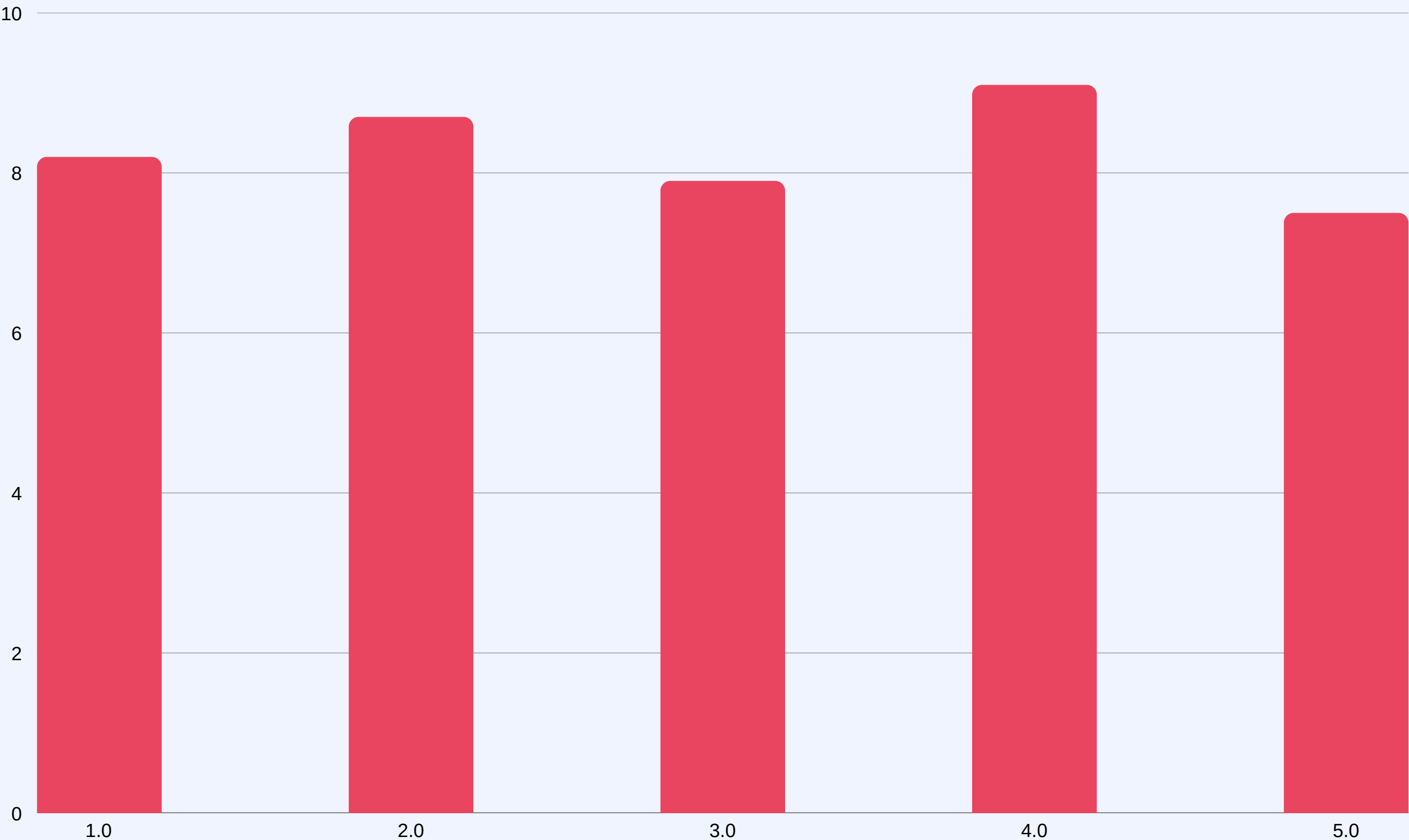
Comparing Prompt Techniques

How each technique performs across key content dimensions

Technique	Best For	Quality	Coherence	Creativity	Complexity
Query Decomposition	Research reports	★★★★	★★★★★	★★★	★★★
Decision-Making	Case studies	★★★★★	★★★★	★★★	★★★★★
Semantic Filtering	Articles / Blogs	★★★★	★★★★	★★★	★★
Structured Prompting	Templates / Docs	★★★★★	★★★★★	★★	★★★★★
Creative Prompting	Stories / Marketing	★★★	★★★	★★★★★	★★

Analyzing Prompt Effectiveness

Quality Score by Technique (Illustrative)



Key Takeaways

- Structured prompting yields highest consistency
- Decision-making prompts produce best analysis
- Creative prompting scores high on originality
- Combining techniques boosts overall quality
- Context specificity improves all scores

Applying Prompt Patterns — Workflow

1

Define Your Goal:

What type of content do you need? Report, article, case study, or creative piece? This determines your technique.

2

Choose a Technique:

Match the content type to the best-fit prompt pattern using the comparison framework.

3

Draft Your Prompt:

Write the prompt using the template: Role → Context → Task → Format → Constraints.

4

Generate & Review:

Run the prompt and critically evaluate quality, coherence, relevance, and creativity.

5

Iterate & Compare:

Try alternate prompt styles on the same task and document differences in output quality.

Key Takeaway

Prompt structure shapes content quality.

- Different prompt patterns produce measurably different outputs
- Structured prompting excels at consistency and professional docs
- Creative prompting unlocks originality and imaginative content
- Combining techniques is often the most powerful approach
- Prompt engineering is a learnable, transferable skill